

William Cook

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EDUCATION

Liberty University

B.S. Computer Science: Software Engineering & Technology Entrepreneurship

Lynchburg, VA

Expected May 2028

- **Relevant Coursework:** Programming in C++ (Advanced), Data Structures & Algorithms, Linux Operating System, Network Architecture & Protocols, Discrete Mathematics

EXPERIENCE

Cole Design & Development LLC

Software Engineering Intern

December 2025 – Present

Hudson, OH

- Extended a Rust/Tauri desktop application's backend to expose protocol-layer commands to a TypeScript/React frontend, enabling full UI-to-hardware communication for an embedded device
- Resolved multiple GitHub issues in production software including configuration persistence, frontend/backend integration failures, and dynamic parameter locking behavior
- Conducted dependency analysis of an embedded C firmware module, mapping all external function and global data dependencies to support future refactoring and architecture review
- Improved simulator reliability by implementing virtual port binding to bypass hardware discovery in simulation mode, reducing connection edge cases during development

Missionize

Full Stack Developer

June 2025 – September 2025

Remote

- Built React Native UIs for habit tracking, journaling, and dashboards for Discipler.io, an AI-powered habit tracking platform for church ministries
- Developed RESTful APIs with FastAPI and MongoDB; managed feature branches and code review via Git

Liberty University Competitive Programming Club

Officer

August 2024 – Present

Lynchburg, VA

- Led team to 6th place out of 61 at the 2024 ICPC Mid-Atlantic Regional; solved 150+ Kattis problems and teaches weekly algorithm lessons in C++ and Python

Software Development Club

Member

January 2025 – Present

Liberty University, Lynchburg, VA

- Placed 2nd at the club's annual hackathon, building a full-featured static website generator in Python and Tkinter under a time constraint
- Participates in weekly hands-on development sessions and speaker events covering modern frameworks, AI tooling, and software engineering best practices

PROJECTS

Self-Hosted Server Infrastructure | *Linux, Raspberry Pi 5, Python, Nginx, systemd, Tailscale*

Ongoing

- Deployed and administers a headless Linux server on a Raspberry Pi 5 (16GB RAM, 1TB NVMe SSD) with SSH key authentication, Tailscale remote access, and automated cron-based backup scheduling across multiple hosted services
- Developed an asynchronous Python background service for real-time log monitoring and external event forwarding, deployed as a lifecycle-bound systemd service; serves a club website on a self-managed Nginx instance with HTTPS

Static Website Generator | *Python, Tkinter, HTML5, CSS3*

2025

- Built a GUI tool for generating responsive static websites with nine themes and live browser preview; achieved 2nd place at the Liberty University Software Development Club Hackathon

TECHNICAL SKILLS

Languages: Python, C (Embedded), C++, Rust, TypeScript, JavaScript, HTML/CSS, SQL

Frameworks & Libraries: React, React Native, FastAPI, Tauri, Node.js, Tkinter, Matplotlib, discord.py

Cloud & Tools: AWS (S3, CLI), Azure (App Service, Cosmos DB), Git, Linux, Nginx, systemd, MongoDB

Concepts: Embedded systems development, full-stack development, REST API design, IoT & MQTT, Linux system administration